

Fall Term 2024

Title:	Assignment 4	Due Date:	09-01-2025 23:59:59
Instructor:	Muhammad Yasin Nasir	Assig Date:	01-01-202
Book:	Assembly Language Programming Organization of the IBM PC by Ytha Yu		
Subject:	Computer Organization & Assembly Language		
Syllabus:	Shifting, Rotation, Stack, Procedures, Multiplication, Division, String		
Name:		Roll No.	

Instructions:

01. Questions are SELF EXPLANATORY. Understanding the Question is part of Solution.
02. Error in Question will be advantageous to Student.
03. Read the Questions carefully before attempting.
04. Attempt All Questions in a Precise Fashion.
05. Manage Your Time. Therefore, there will be no extension in the deadline.
06. Plagiarism may lead to F grade, so be honest.

Submission Method:

01. Assignment must be submitted individually.
02. **Take print out of the given assignment. Use A4 size paper** (210 × 297 millimeters or 8.27 × 11.69 inches) to print it out. **Attach** the outputs of the screen in printed form.
03. Use any scanner tool to convert it to pdf. Text must have clear visibility after scanning. Blurred pdfs will be rejected.
04. Upload the solved assignment (.asm files) at [university portal](#) before the deadline along with asm files. There are two separate threads to submit the asm file for each of both questions.
05. Submit the assignment (hard from) during the first half hour of the subsequent lecture.

Guidelines:

01. Use your official name on the SAP with first letter of each string as capital alphabets and others as small letters.
02. Use your complete SAP ID.
03. Apply **validation** to each input string/number/character.
04. Use the following pattern for file name: sapID_Q1_A3.asm
05. Submit only one file of asm program.
So, you need to assemble all parts of the question in one file.

Q 1: Given the initial values of registers and variables, find the status of the registers and variables after the execution.

Part a: Initial values of registers:

AX=	4040	BX=	01234	CX=	0003	DX=	0002
BP=	0000	SI=	0000	DI=	0000	SP=	0000
ZF=	0	SF=	0	DF=	0	CF=	0

.DATA (starts at 073F:0020 h)

STR1 DB "ABCDE\$"

STR2 DB "FGHIJ\$"

. CODE

.

.

CLD

SCASB

.

.

Values after execution of given instruction:

AX=		BX=		CX=		DX=	
BP=		SI=		DI=		SP=	
ZF=		SF=		DF=		CF=	

Part b: Initial values of registers:

AX=	00FB	BX=	FFFF	CX=	0003	DX=	0002
BP=	0000	SI=	0000	DI=	0000	SP=	0000
ZF=	0	SF=	0	DF=	0	CF=	0

.DATA (starts at 073F:0020 h)

. CODE

.

.

DIV BL

.

.

Values after execution of given instruction:

AX=		BX=		CX=		DX=	
BP=		SI=		DI=		SP=	
ZF=		SF=		DF=		CF=	

Part c: Initial values of registers:

AX=	0ABC	BX=	ABCD	CX=	DC03	DX=	0002
BP=	0000	SI=	0000	DI=	0000	SP=	0000
ZF=	1	SF=	1	DF=	1	CF=	0

.DATA (starts at 073F:0020 h)

. CODE

.
. .
SAR BX, CL

.
Values after execution of given instruction:

AX=		BX=		CX=		DX=	
BP=		SI=		DI=		SP=	
ZF=		SF=		DF=		CF=	

Q 2: Write an assembly language program to concatenate the two strings. You need to write two procedures, first one is STRIN and the other one is STRCONC. Main program should print the output as given in the box.

STRIN: Procedure must have initial address of string as the input. Then Ask the user to input the string of characters terminated by “Enter” key. Save the string at the specified address in memory.

STRCONC: Procedure must have initial addresses of three strings in SI, DI, and BX. Where SI and BX represent the addresses of two source strings and DI represent the address of concatenated string.

Input string1: “Final Term”

Input String2: “Exam”

Final string: “Final Term Exam”

Q 3: For given program, write the output and fill the asked values of registers in the given boxes.

.MODEL SMALL .STACK 100H .DATA VAR1 DW 45H, 56H, 67H, 78H, 89H, 90H .CODE MOV AX,@DATA MOV DS,AX MOV AX,1234H MOV BX,3456H MOV CX,5678H MOV DX,7890H CALL UNKNOWN MOV AH,4CH INT 21H MAIN ENDP UNKNOWN PROC PUSH AX PUSH BX PUSH CX PUSH DX MOV CX,04 @LOOP: MOV AH,02 MOV DL,BYTE PTR VAR1 INT 21H LOOP @LOOP POP AX POP BX POP CX RET UNKNOWN ENDP END MAIN		Write values after the execution.				
		AX	BX	CX	DX	SP
		12 34	00 00	01 3D	00 00	0100
		12 34	34 56	01 3D	00 00	0100
		12 34	34 56	56 78	00 00	0100
		12 34	34 56	56 78	78 90	0100
		AX	BX	CX	DX	SP
		12 34	34 56	56 78	78 90	00 FE
		Fill loop portion for the last iteration.				
		AX	BX	CX	DX	SP
		1234	3456	5678	7890	00FC
		1234	3456	5678	7890	00FA
		1234	3456	5678	7890	00F8
		1234	3456	5678	7890	00F6
		12 34	34 56	0004	78 90	00 F6
		02 45	34 56	00 00	78 45	00F6
		0245	3456	0001	7845	00F6
		0245	3456	0001	7845	Xxxx
		0245	3456	0000	7845	00F6
		7890	3456	0000	7845	00F8
		7890	5678	0000	7845	00FA
		7890	5678	3456	7845	00FC
		78 90	56 78	34 56	78 45	00FE

Q 4: For the given code and its relevant data, find the asked values in the table.

<pre>.MODEL SMALL .STACK 100H .DATA PROMPT_1 DB 'Enter the 16-charcter(0/1) input : \$' PROMPT_2 DB 0DH,0AH,'The output value : \$' .CODE MAIN PROC MOV AX, @DATA MOV DS, AX LEA DX, PROMPT_1 MOV AH, 9 INT 21H XOR BX, BX MOV CX, 16 MOV AH, 1 @LOOP1: INT 21H CMP AL, 0DH JE @END AND AL, 0FH SHL BX, 1 OR BL, AL LOOP @LOOP1 @END: MOV CX, 16 XOR AX, AX @LOOP: ROL BX, 1 JNC @NEXT INC AX</pre>	<pre>@NEXT: LOOP @LOOP MOV CX, AX LEA DX, PROMPT_2 MOV AH, 9 INT 21H MOV AH, 2 CMP CL, 10 JAE @LABEL1 OR CL, 30H MOV DL, CL JMP @DISPLAY @LABEL1: MOV DL, 31H INT 21H SUB CL, 10 OR CL, 30H MOV DL, CL @DISPLAY: INT 21H MOV AH, 4CH INT 21H MAIN ENDP END MAIN</pre>					
Input : 1010101010101010 Output value displayed : 8				Value's of register after execution	AX : 4C 38 BX :AA AA CX: 00 38	DX: 00 38